

Demographic Diversity on the Web

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ABSTRACT

To what extent do the online experiences of, for example, men and women, or Whites and African-Americans differ? We address such questions by pairing web browsing histories for 265,000 anonymized individuals with user-level demographic data—including age, sex, race, education, and income. In one of the most comprehensive analyses of Internet usage patterns to date, we make three broad observations. First, while the majority of popular sites have diverse audiences, there are nonetheless, prominent sites with highly homogeneous user bases. For example, Fox News attracts millions of visitors each month, yet has an audience that is more than 90% White. Second, although most users spend a significant fraction of their time on email, search, and social networking sites, there are still large group-level differences in how that time is distributed. For example, women spend a third more of their time on Facebook than do men, and aside from such universally popular destinations, the top sites frequented by different groups are relatively distinct. Finally, the between-group statistical differences enable reliable inference of an individual's demographic attributes from browsing activity. We thus conclude that while the Internet as seen by different demographic groups is in some regards quite similar, sufficient differences persist so as to facilitate group identification.

Categories and Subject Descriptors

J.4 [Computer Applications]: Social and Behavioral Sciences

General Terms

Human Factors, Measurement

Keywords

demographics, digital divide, online diversity, web browsing

1. INTRODUCTION

The rapid growth and importance of digital technologies over the past several decades has prompted questions regarding access to and use of the Internet by different demographic groups. For example, the “digital divide” [5, 12] between individuals with and without Internet access has been shown to correlate with income, education, and race [6]. Furthermore, even among the growing fraction of those with access to the web, a natural question persists as to whether different demographic groups have disparate online experiences.

A substantial body of survey-based work has attempted to address this question. For example, several authors have found significant race-dependent preferences for social networking sites, with White students preferring Facebook to MySpace [1, 4, 15]. Likewise, work by Jackson et al. [8] suggests that undergraduate women have in the past made more use of e-mail than their male counterparts, though subsequent research fails to find such a gap [9]. Unfortunately, however, survey-based studies offer a limited view of Internet use as they often involve small, non-representative samples of the population, and rely on users' incomplete—and sometimes inaccurate—statements of their own online behavior. Thus, without large scale observational studies, key questions remain unanswered regarding the demographic diversity of the web.

To this end, we explore the web browsing activity of a large, representative population of Internet users. In one of the most comprehensive studies to date, we couple browsing histories for approximately 265,000 anonymized individuals with user-level demographic information—including age, sex, race, education, and income. We are thus able to systematically investigate online diversity with regard to websites, demographic groups, and individuals.

At the site level, we examine the demographic composition of the top 100,000 domains, finding numerous prominent sites with highly homogeneous audiences (Section 4). For example, Petfinder ranks among the top 1,000 most popular sites, drawing millions of users a month, over 90% of whom are White. However, despite the prevalence of such skewed online destinations, a substantial fraction have relatively diverse audiences that mirror the online population at large. In fact, comparing the homogeneity of websites to zip codes, we find that web sites tend to be more racially diverse.

In examining group-level patterns, we focus on how different demographic groups distribute their time online (Section 5). On the one hand, all demographic groups spend

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more than a third of their time on a handful of email, search and social networking sites. On the other hand, there are notable differences in how different groups distribute their time, both on universally popular and on niche sites. For example, while YouTube is heavily frequented by both sexes, men spend more than twice of their time on the site than do women. Likewise, Etsy ranks among the top 100 most popular destinations for women, but is relatively unpopular among men.

Finally, we explore individual-level variation in web activity by quantifying the extent to which browsing history is predictive of a user’s demographic attributes (Section 6). Across all five demographic dimensions that we consider—age, sex, race, education, and income—one can reliably predict individuals’ attributes from the pages that they visit. Thus, while the web as seen by different demographic groups is in some regards quite similar, sufficient differences persist so as to facilitate group identification.

2. RELATED WORK

Aside from the survey work discussed above [1, 4, 8, 9, 15], previous research on patterns of Internet use has generally focused on particular domains or services. Most recently, Chang et al. [2] conducted an extensive analysis of activity on Facebook, showing that the site’s audience has become more ethnically diverse over time, and is converging to a distribution similar to that of the overall online population. They also found that demographic groups tend to differ in their use of site features. For example, Asians are the most engaged group, with high numbers of wall, video, note, gift, comment, and group sharing actions. In other work, Weber and Castillo [16] highlight striking demographic differences in users’ web search intent. For example, when searching for “wagner”, U.S. women are most likely thinking of the 19th century German composer, while men are most likely interested in the paint sprayer company. In a following study, Weber and Jaimes [17] found intriguing differences in how different demographic groups search, and what they search for. Finally, an eye-tracking study by Lorigo et al. [11] likewise found gender differences in search strategies.

In addition to such descriptive work, several studies have focused on predicting demographic attributes from online activity. In analyzing the vulnerability of web search query logs to privacy attacks, Jones et al. [10] found that search queries are highly predictive of sex, age, location, and even individual identity. Similarly, using one week of activity for 189,000 users, Hu et al. [7] were able to reliably infer sex and age from the content of visited sites. Due to the sparse sample of activity used, they first fit a model to predict site-level distributions over demographic dimensions, after which a smoothed, reduced dimensionality Bayesian framework is used to predict user demographics from site visits. In a departure from past predictive work, we note that we are primarily interested in better understanding online demographic diversity; studying individual-level prediction serves as a means to this end.

3. DATA AND METHODS

Our analysis of web browsing behavior is based on complete activity logs for the approximately 265,000 users who participated in the Nielsen MegaPanel between June 2009 and May 2010; on average, panelists spent 138 days in the

study. Each user-generated pageview—which we use as a proxy for time—was weighted by Nielsen to mimic a representative sample of U.S. Internet traffic. In addition to these web browsing histories, individual and household level data were collected from each panelist, including age, sex, race, educational attainment, and household income. For ease of analysis, multiclass and continuous demographic attributes were transformed to binary categories. For example, race was divided into White and non-White, household income was split at \$50,000, and, as per U.S. Census conventions, individuals over the age of 25 were categorized as adults. We note that all data were anonymized to protect the privacy of participants.

The web addresses for visited sites were normalized by removing any leading ‘www’ and then retaining at most three domain levels. For example, `www.yahoo.com` was transformed to `yahoo.com`, and `us.mg0.mail.yahoo.com` was converted to `mail.yahoo.com`. In total, 9.2 million normalized domains were viewed by at least one panelist; we limit our analysis to the 100,000 sites viewed by the most users. After this restriction, there were in aggregate 3.4 billion pageviews, with a median of 3,700 pageviews per user.

Given the size of the dataset, data manipulation was carried out with Hadoop and Pig [13], open source tools for scalable, distributed computing. To infer demographic attributes, we fit models with SVM^{light} [14], as described in more detail in Section 6. Descriptive statistical analysis was conducted in R, and graphs were generated with ggplot2 [18].

4. THE DEMOGRAPHIC COMPOSITION OF WEB SITES

To what extent do online destinations attract homogeneous audiences—audiences that are, for example, almost exclusively female, or White, or college educated? Along each of the demographic dimensions we consider—sex, race, education, age, and income—we find not only that highly homogeneous sites exist, but that many of these sites are quite popular, attracting hundreds of thousands or even millions of visitors per month. Despite the existence of such highly skewed sites, however, a sizable fraction of sites are relatively diverse.

For the 100,000 websites in our dataset, we first compute their skew along several demographic dimensions. Specifically, we calculate the proportion of each site’s pageviews that are generated by women, Whites, college educated users, adults (i.e., users over 25), and users with household incomes less than \$50,000.¹ As shown in Figure 1, we find that a substantial fraction of domains have demographic compositions similar to the overall online population. For example, approximately half of all websites have audiences that are between 65% and 85% White (the online population average is 77%), and half have audiences that are between 65% and 85% college educated (the online average is 75%). Site skew is more pronounced for sex, but even along this dimension we find that approximately one-third of sites have audiences between 55% and 75% female, relatively close to the online average of 66%.

¹In computing a site’s education skew, we first restrict to users who are at least 25 years old, and then consider the percentage of these users that have at least some college education.

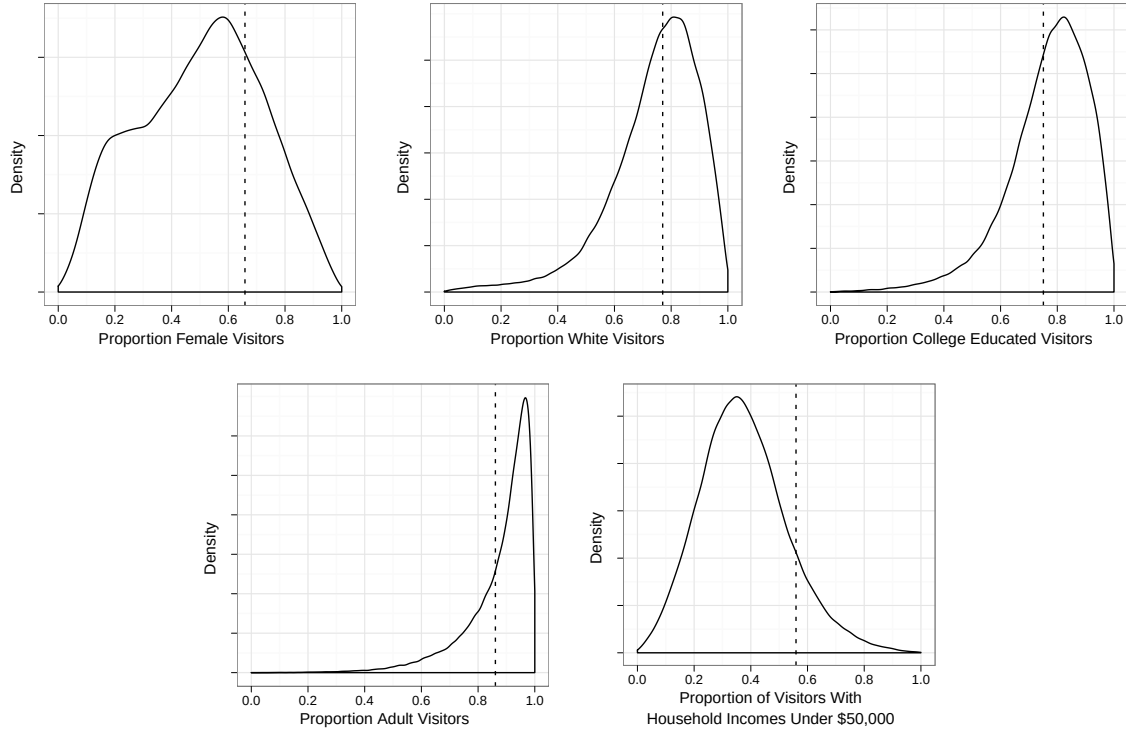


Figure 1: Distributions of site skew over all 100,000 sites in the dataset. The dotted lines show estimated means over the U.S. Internet population. In computing summary statistics, users are weighted by pageviews.

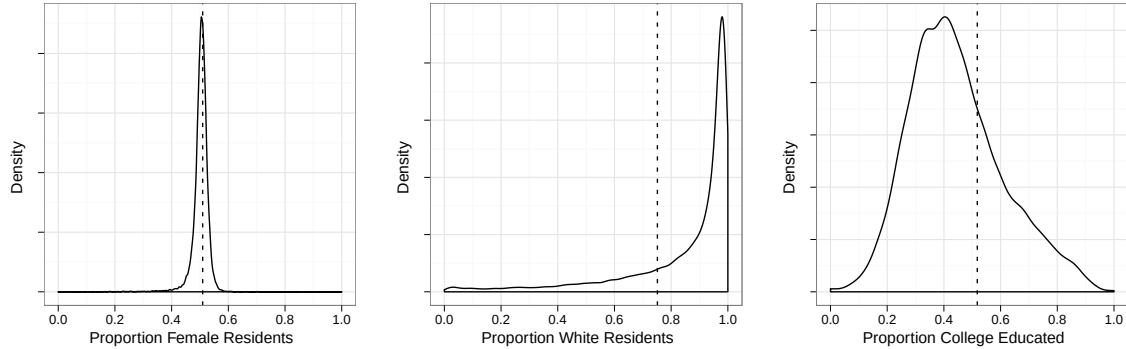


Figure 2: Distributions of residential demographic skew by zip code.

As a point of comparison, we contrast the sex, race and education composition of websites to that of offline neighborhoods based on data compiled by the U.S. Census Bureau. We find that while sites exhibit more sex homogeneity, they tend to be more racially diverse than neighborhoods. As one would expect, and as confirmed by Figure 2, zip codes are by and large gender balanced, with 95% of zip codes composed of between 45% and 55% women. By comparison, 20% of sites have audiences that are at least 80% female or 80% male. Thus, relative to zip codes, websites exhibit pronounced sex homogeneity. With regard to race homogeneity, however, we find the opposite effect: web site audiences are typically more diverse than neighborhoods. Among zip codes, the median proportion of White residents

is 94%, compared to 77% of audience members among web sites. Finally, we observe, at least qualitatively, that both websites and zip codes have similar levels of education heterogeneity relative to the online (75%) and offline (52%) population averages; the higher proportion of college educated Internet users complicates more rigorous comparison.

While a substantial fraction of websites have relatively diverse audiences, we also note the existence of extremely skewed sites, as depicted by the heavy tails of the site skew distributions (Figure 1). Specifically, among the complete set of 100,000 domains, we find sites with audiences that are more than 95% skewed toward each demographic group (e.g., sites that are more than 95% male, female, White, and non-White, etc.). In fact, for many of these demographic at-

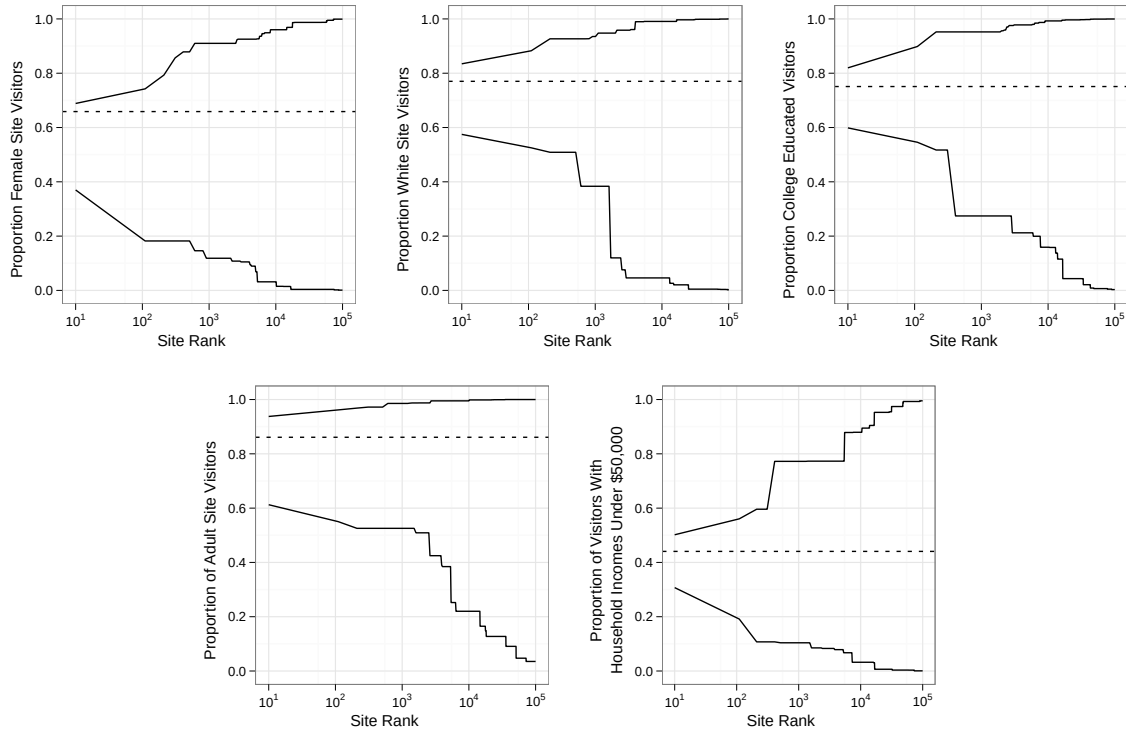


Figure 3: Range of site skew as a function of popularity. For example, among the top ten sites, the proportion of female site visitors ranges from approximately 38% to 68%; however, among all 100,000 sites that we examine, site skew ranges from 0% female to 100% female. The dotted lines show estimated means over the U.S. Internet population. In computing summary statistics, users are weighted by pageviews.

tributes, there are sites that are more than 99% skewed. In other words, even among these relatively popular 100,000 sites—the least popular of which still garners thousands of visitors per month—one finds extremely homogeneous destinations along all of the demographic dimensions we consider.

While the top 100,000 sites already represent the most frequented online destinations in a universe of millions of websites, we note that this skew persists even as one restricts attention to increasingly popular destinations. Figure 3, which shows the range of site skew as a function of site popularity, illustrates this point, revealing highly homogeneous sites even among the top 10,000 most popular domains. For example, the news site `foxnews.com` and the social networking site `blackplanet.com`—both of which attract millions of users each month—have audiences that are more than 90% White and non-White, respectively.² A selection of other popular sites with more than 90% skew is shown in Table 1. Among the most popular female dominated destinations, for example, are the shopping-related sites `youravon.com` and `collectionsetc.com`. Conversely, the sports site `coveritlive.com` and the adult entertainment site `needlive.com` are among the most popular male dominated sites. For education, we find `news.google.com` to be one of the most popular sites with a predominately college

educated audience; and for income, `opentable.com`—a real-time restaurant reservation service—and `marketwatch.com` are popular sites frequented largely by relatively wealthy users.

Although none of the top ten domains exhibit extreme demographic skew, Figure 3 shows that there are still substantial differences between these sites. For example, consistent with past survey work [1, 4, 15], the social networking sites Facebook and MySpace are approximately 75% and 60% White, respectively—a large difference for two highly popular sites that provide similar services. Likewise for income, while approximately 50% of pageviews on MySpace are generated by households with incomes less than \$50,000, only 35% are generated by such households on Facebook.

5. THE WEB BROWSING BEHAVIOR OF DEMOGRAPHIC GROUPS

Whereas in Section 4 we examined the demographic composition of web sites, here we consider the aggregate web browsing behavior of different demographic groups. Transitioning from a site-centric to a group-centric analysis, we focus on where groups spend their time online and how that time is distributed. On the one hand, regardless of demographics, users tend to concentrate their time on a core set of email, search, and social networking sites. On the other hand, however, there is substantial variation in how different groups distribute their time on these core sites, and also in which niche sites they frequent.

²Site-conditional demographic skew should not be conflated with the frequency of site visits conditioned on demographics (i.e., $p(y|x) \neq p(x|y)$). For instance, while 93% of pageviews on Fox News are generated by White users, only a small fraction (0.1%) of pageviews by White users are on Fox News.

	Greater Than 90%	Less Than 10%
Female	youravon.com collectionsetc.com	coveritlive.com needlive.com
White	foxnews.com wunderground.com	blackplanet.com mediatakeout.com
College Educated	news.google.com nytimes.com	slumz.borden.com sythe.com
Over 25 Years Old	mail.yahoo.com apps.facebook.com	nanowrimo.org cbox.ws
Household Income Under \$50,000	scarleteen.com boards.adultswim.com	opentable.com marketwatch.com

Table 1: A selection of popular sites that are homogeneous along various demographic dimensions.

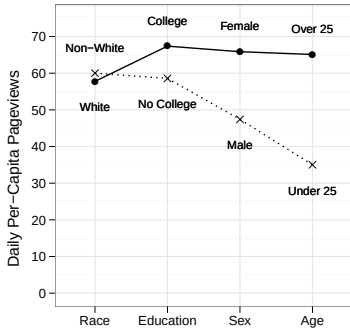


Figure 4: Average daily per-capita pageviews by demographic group. The solid line represents majority groups and the dashed line represents minorities.

We first note that there are large group-level differences in aggregate web browsing volume. For example, as shown in Figure 4, while men on average generate 47 pageviews per day, women generate 66 per day—40% more. Education differences are less pronounced but still substantial, with college educated adults generating approximately 15% more pageviews (67 per day) than those who did not attend college (59 per day). We do not see noticeable race difference, with both Whites and non-Whites exhibiting approximately the same per-capita usage.

To capture the subtleties of browsing behavior in more detail, we move from examining aggregate volume to per-site usage of the top fifty most popular domains—destinations that in total account for approximately 45% of all web browsing. Figure 5 plots the percent of time—on a log scale—that each demographic group spends on these top domains. For example, Facebook is by far the most frequented online destination by both men (who spend 6% of their time there) and by women (8%). As another example, adults, women, Whites, college educated users, and relatively wealthy individuals all spend significantly less—sometimes less than half as much—of their time on YouTube than their demographic counterparts.

These results illustrate two broad points. First, all demographic groups spend a large fraction (over 30%) of their time on portals such as Yahoo!, services such as Google, and social networking sites such as Facebook. Second, as is

visually apparent from Figure 5, there are significant differences in how groups distribute their time on the web. These differences—which, as mentioned above, hold for highly frequented sites such as Facebook and YouTube—are in some cases even more pronounced for lower traffic sites. For instance, the gaming site `pogo.com` accounts for less than 1% of pageviews among both low and high income users, but low income users spend almost twice as much of their time there.

To further explore group-level differences in browsing behavior, we switch focus from the overall most popular sites to the top sites visited by different demographic groups. For example, looking at the top fifty sites visited by women and men, we find that a substantial number of the top sites for women (e.g., `walmart.com`) do not make the list for men; conversely, many top sites for men (e.g., `sports.yahoo.com` and `finance.yahoo.com`) are absent from the female top fifty. To more precisely characterize the similarity between the top female and top male sites, we compute the Jaccard index between the two lists, finding that 61% of the top sites make the lists for both groups.³ Repeating this analysis along the other demographic dimensions—race, education, age, and income—we observe comparable Jaccard similarity (Figure 6). For example, there is 64% overlap in domains on the top fifty White and non-White lists—with, for instance, `mapquest.com` and the anime-themed site `gaiaonline.com` appearing on only the White and non-White top fifty, respectively. Thus, while we find substantial overlap among the most frequented sites by different demographic groups, we also observe reasonably large group-level variation.

6. INFERRING DEMOGRAPHICS FROM WEB BROWSING HISTORIES

Though we have above noted substantial group-level differences in browsing behavior, a question remains whether such differences are observable at the level of individuals. Here we explore this possibility in detail, examining the degree to which demographics can be predicted by browsing histories, the value of popular sites for predictive quality, and the variation in performance across users.

For each of the five demographic dimensions—sex, age, race, education, and household income—we formulate the prediction task as binary classification (e.g., distinguishing

³The Jaccard index between two sets A and B is defined as $J(A, B) = |A \cap B| / |A \cup B|$.

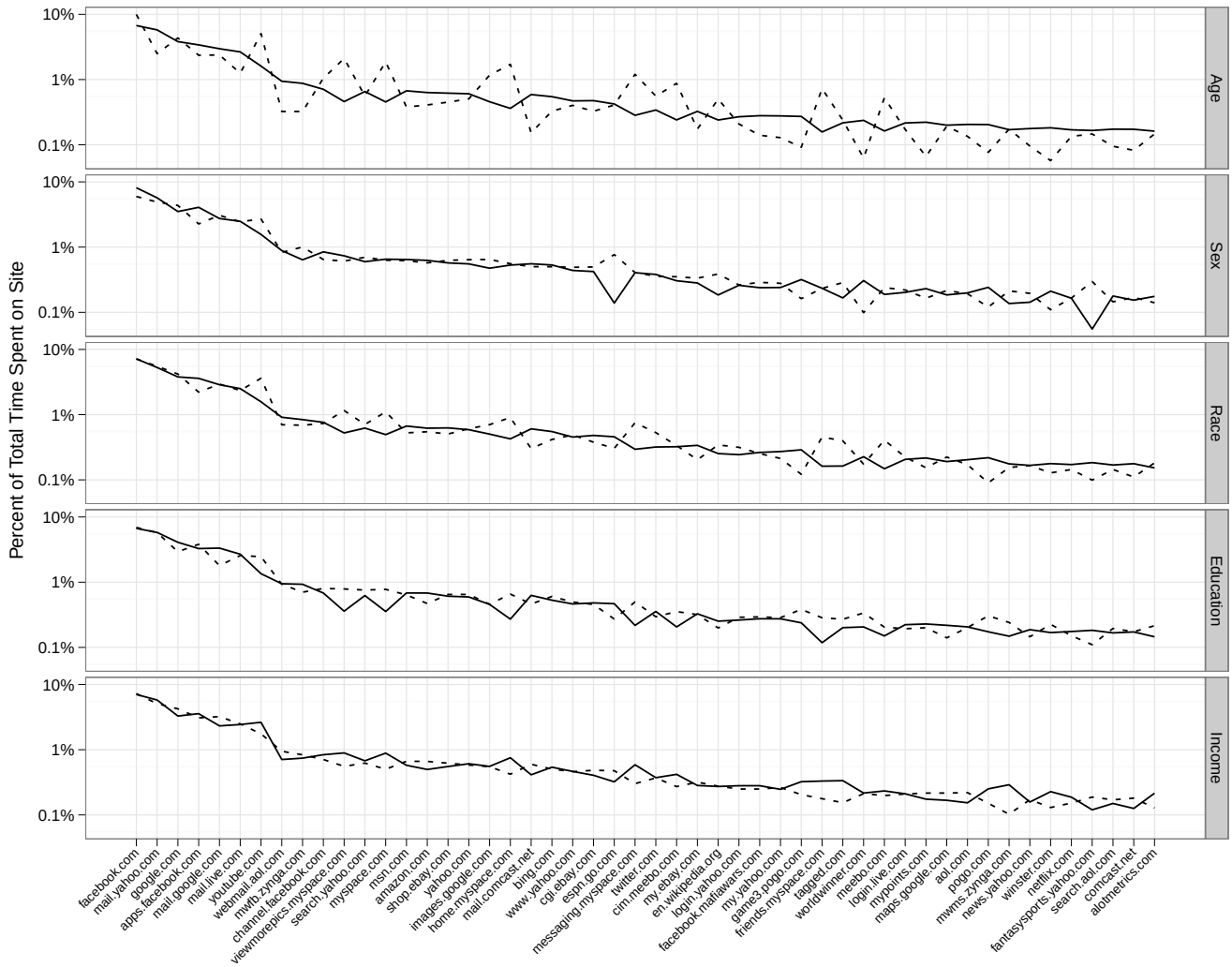


Figure 5: Proportion of time spent by different demographic groups on the top fifty domains. Solid lines represent majority groups (adults, women, Whites, college educated users, and users with household incomes greater than \$50,000); dashed lines correspond to minority groups (non-adults, men, non-Whites, non-college educated users, and those with household incomes less than \$50,000).

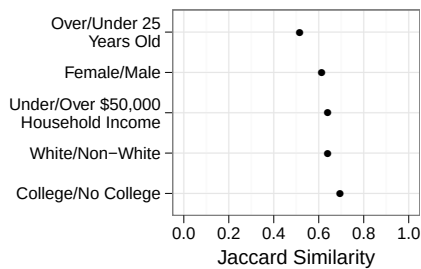


Figure 6: Jaccard similarity between the top 50 sites visited by different demographic groups.

between females and males, or Whites and non-Whites). In addition to the 100,000 normalized domains described above, we include aggregated domains as additional predictors (e.g., `mail.yahoo.com` and `sports.yahoo.com` are collapsed to `*.yahoo.com`), resulting in approximately 114,000 total features. In predicting age, sex, race, and education, users are represented by sparse binary vectors indicating which domains (or aggregated domains) they visited during the study period; for household income, we consider the set of domains visited by the entire household.⁴

Given the scale of the data, and that the number of features (114,000) is comparable to the number of examples (265,000), we use linear support vector machines (SVMs) [3] to infer demographic attributes from browsing history. Lin-

⁴Variants of these features, for example visitation frequency or tf-idf weights, result in nearly identical performance.

	Large positive weight	Large negative weight
Female	winster.com lancome-usa.com	sports.yahoo.com espn.go.com
White	marlboro.com cmt.com	mediatakeout.com bet.com
College Educated	news.yahoo.com linkedin.com	youtube.com myspace.com
Over 25 Years Old	evite.com classmates.com	addictinggames.com youtube.com
Household Income Under \$50,000	eharmony.com tracfone.com	rownine.com matrixdirect.com

Table 2: A selection of the most predictive (i.e., most highly weighted) sites for each classification task.

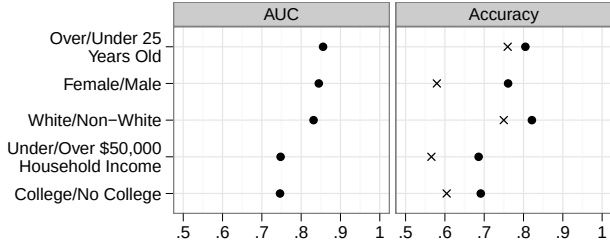


Figure 7: Summary of model performance, indicated by solid circles, for all demographic attributes. Population skew is given by x’s for comparison. Note that higher AUC closely corresponds to lower Jaccard similarity, as shown in Figure 6.

ear SVMs generate predictions of the form

$$\hat{y}(x_i) = w \cdot x_i + b$$

where the predicted class is defined by the sign of $\hat{y}(x_i)$. To guard against overfitting, SVMs seek the weight vector w that maximally separates the positive and negative examples in the training set. Specifically, SVMs optimize the loss function

$$L(y, \hat{y}) = C \sum_i [1 - y_i \hat{y}(x_i)]_+ + ||w||^2$$

where $[x]_+ = (|x| + x)/2$ indicates the positive part, and C is a tunable parameter that balances model fit against generalization. Users are randomly divided into an 80% training set on which models are fit, a 10% validation set used to select the optimal parameter C for each demographic attribute, and a 10% held-out test set on which we evaluate and report final performance.

Figure 7 summarizes our results for all five classification tasks. The right panel displays the accuracy of predictions, showing reasonable performance across all demographic dimensions, with slightly higher accuracies for age, sex, and race—80%, 76%, and 82%, respectively—than for education and income—70% and 68%. To help put these numbers in perspective, Figure 7 also includes the overall population skew for each demographic attribute, indicated by x’s (e.g., 57% of the online population is female, while 76% is comprised of adults).

Given the substantial demographic skew, we also present AUC—or area under the ROC curve—in the left panel of

Figure 7, a measure that effectively re-normalizes the majority and minority classes to have equal size. Intuitively, AUC is the probability that a model scores a randomly selected positive example higher than a randomly selected negative one (e.g., the probability that the model correctly distinguishes between a randomly selected female and male). Though an uninformative rule would correctly discriminate between such pairs 50% of the time, predictions based on browsing histories are relatively reliable, ranging from 74% to 85%. Thus, whether we measure performance in terms of accuracy or AUC, we find that browsing activity provides a strong signal for inferring individual-level demographic attributes.

A benefit of linear models is the interpretability of the weight vector w . In Table 2, we report a sample of the most predictive (i.e., largest positively and negatively weighted) sites for each attribute. For example, visiting the popular cosmetics company [lancome-usa.com](#) strongly indicates that a user is female, while visits to the sports sites [sports.yahoo.com](#) or [espn.go.com](#) are highly predictive of being male. Interestingly, and perhaps less apparent, the collaborative gaming community site [winster.com](#) is also among the highest weighted female-predictive sites; closer inspection reveals that the site was created by a northern California housewife as an alternative to gaming destinations that cater to young males. Analogously, visits to Country Music Television ([cmt.com](#)) are a strong indicator of being White, while visits to Black Entertainment Television ([bet.com](#)) are a strong non-White indicator. Though visits to highly weighted sites provide strong cues, we note that many such sites are frequented by a relatively small fraction of the population. Thus, model performance is likely enhanced by the many weak signals from visits to popular but less discriminating sites.

We next examine whether demographic differences in on-line activity—as measured by predictive quality—persist as we restrict to increasingly popular sites. As shown in Figure 8, models fit on as few as the top 1,000 sites perform only marginally worse than those fit on all 114,000 domains (far right)—in other words, even on these top sites, demographic differences are relatively large. For example, in predicting sex using the top 1,000 sites, AUC decreases only four percentage points, from 75% to 71%. That visits to popular—and relatively heterogenous—sites are quite informative is a testament to the aggregate strength of weak signals.

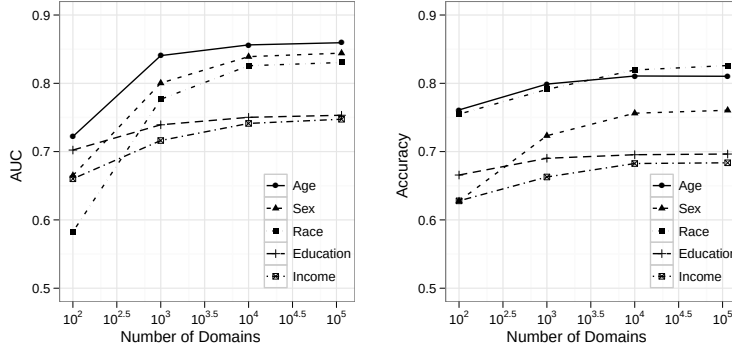


Figure 8: AUC and accuracy as the number of domains used in the predictive models is varied. The far right of each plot indicates all 114,000 features, with points to the left restricted to increasingly popular domains.

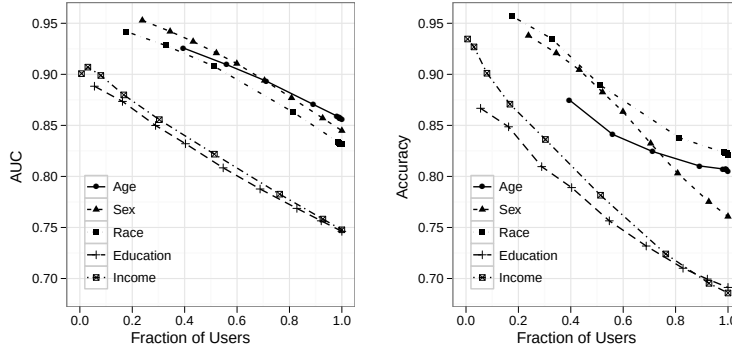


Figure 9: Change in model performance as the fraction of users on which predictions are made is varied. The rightmost points include predictions for all users, whereas points to the left are restricted to users farther from the decision boundary.

Finally, we investigate the extent to which users are more or less stereotypical—under the predictive model—of their demographic group. Intuitively, those users that exclusively visit sports sites should be easier to classify as male than those who visit both sports and cosmetics sites. In quantitative terms, by restricting predictions to users for which the model is most confident, we expect gains in performance. Figure 9 illustrates this tradeoff between performance and coverage as we restrict to increasingly smaller sets of users that lie far from the decision boundary. For example, by restricting to the 50% percent of the population for which the model is most confident, we achieve AUC of approximately 0.9 for sex, race, and age, and approximately 0.8 for education and income—a roughly five percentage point increase compared to predictions over the entire population.

In practice, ad networks, by recording the sites on which users see advertisements, could generate samples of browsing history, and might thus benefit from the demographic predictions we discuss above. We emphasize, however, that such potential applications are neither our aim nor are advocated.⁵ Rather, our objective is to investigate demographic differences through the lens of prediction. The relatively

accurate predictions we report demonstrate that group-level demographic differences are typically larger than individual-level variation.

7. DISCUSSION

Returning to the motivating question—how do the online experiences of demographic groups vary?—we find two answers. A large fraction of sites—especially among the most popular—have relatively diverse audiences that mirror the online population at large, and moreover, all demographic groups engage in the same core activities of email, web search, and social networking. However, substantial group-level differences persist, as noted by the prevalence of prominent, highly homogeneous online destinations, and by variation in how groups distribute their time on core and niche sites alike. In particular, these differences are sufficiently pronounced that one can reliably predict an individual’s demographic attributes from the websites he or she visits. Thus, as we find considerable similarities as well as differences, whether demographic groups have disparate views of the web is largely a matter of perspective.

Such findings are subject to a key limitation, namely that we have investigated online diversity primarily in terms of site visits. In particular, we have not directly examined the content of sites, nor have we considered behavior within given domains (e.g., differences in user activity on Facebook,

⁵For such applications, it is likely that performance could be improved by incorporating additional information, for example geographic-level demographics inferred from a user’s IP address, as available from api.infochimps.org.

as discussed by Chang et al. [2]). Furthermore, while necessary for meaningful analysis, normalizing to 3-level domains has the effect of both aggregating sites that may be qualitatively distinct, and conversely, uncoupling sites that are substantively identical. While addressing these concerns might be valuable, we feel the simplicity of our approach nevertheless lends itself to interpretability and provides insight into the matters at hand.

Finally, though we have attempted to identify and characterize patterns of demographic diversity on the web, the social and policy implications of our findings remain open to debate. Whether the differences we report are cause for alarm or simply reflect individual preferences is a complex question, and one that we leave largely unanswered. We hope, however, that our approach and findings help to clarify and to further discussions of online diversity.

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